

# ESMI Applied Mathematics Project Syllabus

## Queueing Theory

10 June – 8 August 2023

**Instructor:** Laurel Newman (lan66@cornell.edu)

**Lecture Location:** B30B Mann Library

**Lecture Time:**

- Weeks 1 & 2: 9:00-12:00 M\T\W\R\F
- Weeks 3-9: 1:00-3:00 M\W\F

**Office Hours:** By appointment

## 1 Introduction

This summer we will gain experience in doing mathematical research and cover a brief introduction to queueing theory. We will cover probability, some relevant statistical distributions, and Python. Students will work in small research groups to develop their own question about a real-world queueing system and use mathematical and computational tools to gain insight into the system. Students will write up their results and present them at the end of the program.

### 1.1 Resources

Here are some additional resources you may wish to make use of if you wish to explore a topic in more depth than we cover in lecture.

Some further reading on Queueing Theory: [https://www.cs.ucf.edu/~lboloni/Teaching/EEL6785\\_Fall2010/slides/QueueingTheory.pdf](https://www.cs.ucf.edu/~lboloni/Teaching/EEL6785_Fall2010/slides/QueueingTheory.pdf)

NumPy for beginners: [https://numpy.org/doc/stable/user/absolute\\_beginners.html](https://numpy.org/doc/stable/user/absolute_beginners.html)

Introductory probability: <https://math.dartmouth.edu/~prob/prob/prob.pdf>

A great resource for brainstorming queueing problems: <https://opexlearning.com/resources/queueing-theory/>

### 1.2 Timeline

1. Monday 10 June: program starts, beginning first two weeks of daily lectures

2. Tuesday 24 June: students' summer courses begin, we now meet M/W/F.
  - Mondays: skills and best practices
  - Wednesdays: continuing lectures/work days
  - Fridays: work days
3. Friday 28 June: students finalize their research groups and submit their project proposals and group contracts
4. Friday 26 July: students submit their draft research report, poster, and presentation
5. Friday 2 August: "working draft" of final research report, poster and presentation due
6. Wednesday 7 August: students submit their final research report, poster, and presentation
7. Thursday 8 August: Summer Research Symposium

### **1.3 Schedule**

| Week | Day      | Lecture                                             |
|------|----------|-----------------------------------------------------|
| 1    | 10 June  | Introduction                                        |
|      | 11 June  | Intro to Queues                                     |
|      | 12 June  | Intro to Prob                                       |
|      | 13 June  | Prob II: Independence                               |
|      | 14 June  | Prob III: Useful distributions                      |
| 2    | 17 June  | Queues with Randomness                              |
|      | 18 June  | Intro to Python                                     |
|      | 19 June  | Juneteenth                                          |
|      | 20 June  | Python II                                           |
|      | 21 June  | NumPy for Queueing Theory                           |
| 3    | 24 June  | Intro to Research                                   |
|      | 26 June  | Graphing your results                               |
|      | 28 June  | Project Proposals & Group Contracts Due             |
| 4    | 1 July   | Coding Documentation                                |
|      | 3 July   | Project Planning                                    |
|      | 5 July   | Work Day                                            |
| 5    | 8 July   | How to write up your results                        |
|      | 10 July  |                                                     |
|      | 12 July  | Work Day                                            |
| 6    | 15 July  | Effective Math Communication                        |
|      | 17 July  |                                                     |
|      | 19 July  | Work Day                                            |
| 7    | 22 July  |                                                     |
|      | 24 July  |                                                     |
|      | 26 July  | Poster, presentation, report draft due              |
| 8    | 29 July  |                                                     |
|      | 31 July  |                                                     |
|      | 2 August | “Working draft” of poster, presentation, report due |
| 9    | 5 August | Presentation Prep                                   |
|      | 7 August | Presentation Prep & final deliverables due          |
|      | 8 August | Presentations                                       |